

Single Source Justification (Simplified Acquisition)**Section A: General Contract Information**Is this a Bridge Action as defined in the [AF Bridge Action Reduction Plan?](#)☐ Yes☒ No

Contracting Activity: AFTC/PZZF

Project/Program Name: Aerial Refueling Aircraft Simulator Qualification Purchase

Request / Local ID Number: F1S0AY5104A001 / ACQ-25-H72

Estimated Contract Cost (including Options): \$ [REDACTED]

Section B: Description of the Supplies/Services Required

The item to be acquired is a load cell transducer capable of measuring torque and a directional force, otherwise known as a Biaxial torque and force sensor. The sensor will be mounted to the pilot control column, a very high standard for safety and reliability is presented, and will be heavily scrutinized during design review.

The following brand name Futek items are required:

Qty 3 - Item #: QSH02089 pancake style biaxial thrust and torque sensor

Qty 3 - Item #: SLB00002 NIST traceable load certification

Qty 3 - Item #: STB00002 NIST traceable torque certification

Section C: Justification for Soliciting from a Single Source or Brand Name

Project ARASQ has a requirement to simultaneously measure pilot input forces and torques imparted on the control yoke during flight test, with the intention to recreate similar forces in a simulator. The engineering team is tasked with selecting a sensor that meets, or exceeds, the requirements and can minimize any risk associated with modifying crucial aircraft control mechanisms.

Concerns over safety are very high with this type of instrumentation, as a failure could render the aircraft inoperable. Additionally, the KC-46 airworthiness process more heavily scrutinizes our design choices due to FAA involvement. For these reasons the sensor needs to be the least intrusive as possible to the aircraft, implementing only sensors that are specifically designed for this purpose, with a proven record of safety and reliability.

During research for a suitable item, it was concluded that there is only one vendor who creates such a sensor. The item has been used for the same purposes on other aircraft platforms with success.

Without these particular sensors, we would incur a significant amount of design effort and time to build something of our own creation which would have a significant impact to ARASQ scheduled modification and execution timelines. Our timelines are already extremely complicated to organize because they have to be coordinated between other aircraft platforms, CTF's, and other bases.

Acquiring this item significantly reduces the design efforts and mitigates safety risk, allowing ARASQ schedules to remain on track.

Section D: Efforts to Obtain Competition

Information was requested from 5 different vendors who are market leaders in load cell force measurements. The vendor's product teams were consulted to find a product that meets all of electrical, mechanical, and safety requirements. Requests for quotes were sent to the five vendors for pricing and lead times. Once all quotes were received, FUTEK was the only company that could provide a technically acceptable product in the time frame that meets project needs.

Section E: Steps to Preclude Future Single Source or Brand Name Awards

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Market survey and research was performed to find the capabilities of all vendors that provide similar products. Project ARASQ engineers attended several trade shows throughout the year to identify potential vendors or organizations that could provide the required sensors. Engineers will continue to browse the internet and attend trade shows to continue searching for potential vendors.

Section F: Contracting Officer's Determination

I have determined, in accordance with [FAR 13.106-1\(b\)\(1\)](#), that the circumstances of this contract action deem only a single source is reasonably available.

Date	Typed Name and Rank/Grade of Contracting Officer	Signature of Contracting Officer
21-Apr-2025	Christina Day, NH-III	